

Practical No. 9

9. Using a numpy module create an array and check the following:

1. Type of array 2. Axes of array 3. Shape of array 4. Type of elements in array

```
import numpy as np
# Creating array object
arr = np.array([[1, 2, 3],[4, 2, 5]])

# Printing type of arr object
print("Array is of type: ", type(arr))

# Printing array dimensions (axes)
print("No. of dimensions: ", arr.ndim)

# Printing shape of array
print("Shape of array: ", arr.shape)

# Printing size (total number of elements) of array
print("Size of array: ", arr.size)

# Printing type of elements in array
print("Array stores elements of type: ", arr.dtype)
```

Output:

```
Array is of type:  <class 'numpy.ndarray'>
No. of dimensions:  2
Shape of array:  (2, 3)
Size of array:  6
Array stores elements of type:  int64
```

Process finished with exit code 0

Practical No. 12

12. Write a Python program to find first n prime numbers.

```
num = 11
# If given number is greater than 1
if num > 1:
    # Iterate from 2 to n // 2
    for i in range(2, (num//2)+1):
        # If num is divisible by any number between
        # 2 and n / 2, it is not prime
        if (num % i) == 0:
            print(num, "is not a prime number")
            break
    else:
        print(num, "is a prime number")
else:
    print(num, "is not a prime number")
```

Output:-

```
11 is a prime number
```

```
Process finished with exit code 0
```

Practical No. 13

13. Write a Python Program to multiply matrices

```
import numpy as np
```

```
A = np.array([
    [12, 7, 3],
    [4, 5, 6],
    [7, 8, 9]
])
```

```
B = np.array([
    [5, 8, 1, 2],
    [6, 7, 3, 0],
    [4, 5, 9, 1]
])
```

```
# Initialize a 3x4 result matrix filled with zeros
```

```
# (A 3x3 multiplied by a 3x4 results in a 3x4 matrix)
```

```
result = np.zeros((3, 4), dtype=int)
```

```
for i in range(len(A)):
```

```
    for j in range(len(B[0])):
```

```
        for k in range(len(B)):
```

```
            result[i, j] += A[i, k] * B[k, j]
```

```
# Print the final result
```

```
print("Resulting 3x4 Matrix:")
```

```
print(result)
```

Output:-

```
Resulting 3x4 Matrix:
```

```
[[114 160 60 27]
```

```
 [ 74  97  73 14]
```

```
 [119 157 112 23]]
```

```
Process finished with exit code 0
```

Practical No. 14

14. Write a Python program to find the exponentiation of a number.

```
num=int(input("Enter number: "))
exp=int(input("Enter exponential value: "))
result=1
for i in range(1,exp+1):
    result=result*num
print("Result is:",result)
```

Output:-

```
Enter number: 5
Enter exponential value: 3
Result is: 125
```

```
Process finished with exit code 0
```

Practical No. 15

15. Using a for loop, write a program that prints out the decimal equivalents of $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{10}$

```
a = 1
sum = 0
# Running for loop from 2 to 10
for b in range(2, 11):
    c = a / b
    print(f"{a}/{b} :", c)
    sum = sum + c
print(f" $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{10} =$ ", sum)
```

Output:-

```
1/2 : 0.5
1/3 : 0.3333333333333333
1/4 : 0.25
1/5 : 0.2
1/6 : 0.16666666666666666
1/7 : 0.14285714285714285
1/8 : 0.125
1/9 : 0.11111111111111111
1/10 : 0.1
 $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{10} =$  1.928968253968254
```

Process finished with exit code 0

Practical No. 10

10. Write a python program to concatenate the dataframes with two different objects.

```
import pandas as pd
import numpy as np

df1 = pd.DataFrame(np.random.randint(25, size=(4, 4)),
                    index=["1", "2", "3", "4"],
                    columns=["A", "B", "C", "D"])

df2 = pd.DataFrame(np.random.randint(25, size=(6, 4)),
                    index=["5", "6", "7", "8", "9", "10"],
                    columns=["A", "B", "C", "D"])

# concatenating df1 and df2 along rows
vertical_concat = pd.concat([df1, df2], axis=0)

print(vertical_concat)
```

Output:-

	A	B	C	D
1	23	15	14	10
2	14	16	24	16
3	14	24	9	14
4	6	0	13	1
5	0	9	6	9
6	14	5	19	5
7	21	16	4	19
8	16	16	16	5
9	13	15	23	9
10	3	18	18	17

Process finished with exit code 0

Practical No. 11

11. Write a python code to set background color and pic and draw a circle using turtle module.

```
import turtle
```

```
# Get input from the user first
```

```
r = int(input("Enter the radius of the circle: "))
```

```
col = input("Enter the color name or hex value (e.g., #FF0000): ")
```

```
# Open the turtle window after getting input
```

```
screen = turtle.Screen()
```

```
t = turtle.Turtle()
```

```
# Set the fill color
```

```
t.fillcolor(col)
```

```
# Draw and fill the circle
```

```
t.begin_fill()
```

```
t.circle(r)
```

```
t.end_fill()
```

```
# Keep the window open
```

```
turtle.done()
```

```
Enter the radius of the circle: 120
```

```
Enter the color name or hex value (e.g., #FF0000): #25782C
```

Python Turtle Graphics

